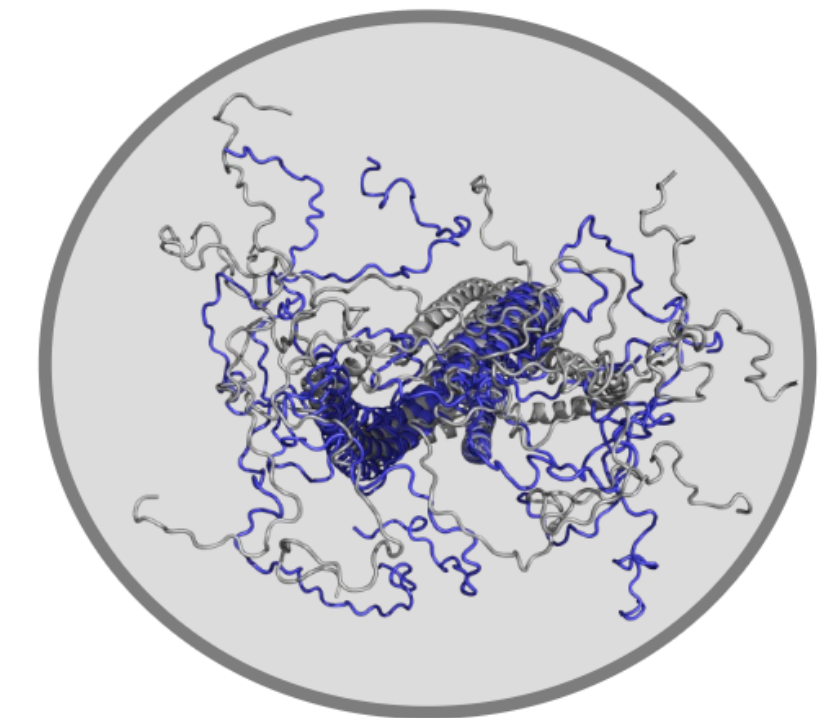


Biological datasets for computational physics

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Biological datasets for computational physics

Biopython practical

Installing biopython

Working locally? Google Colab?

- In order to work locally, you will need to have at least a package manager (conda or miniconda), in order to create an environment and install the required libraries. The steps are as follows:
 1. Create a virtual environment
 2. Activate the environment
 3. Install Jupyter Notebook, matplotlib and biopython
- Otherwise, you can work remotely on Google Colab:
 1. Install matplotlib and biopython
 2. Upload the files necessary for the practical

Recommendations

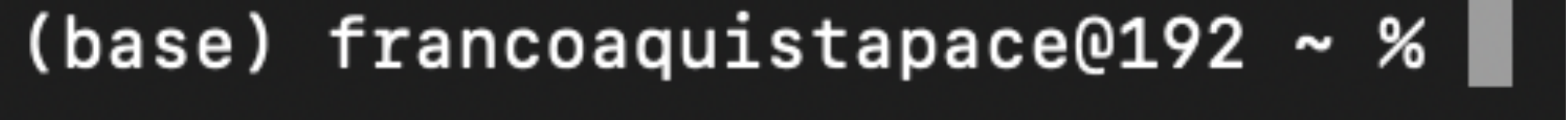
- If you are already familiar with creating and managing virtual environments, I would suggest to run the practical locally:
 - **Benefit:** not needing to re-upload the data every time you access the notebook
- However, if you do not feel comfortable with environment setup, then Google Colab is a nice alternative:
 - **Benefit:** not needing to manage the environment, just running one installation line is enough

(Learning to manage environments will be generally useful for other courses / anything related to python, so if you want to try I will help you)

How to set up your local environment

Disclaimer: we will not go over how to install conda, so you should have it installed beforehand to follow this part

- Make sure that conda is initialized:
 - You should see something like this
 - Otherwise, run: `conda init`
- Now we can create a new environment and install the packages:
 - Run: `conda create --name bio_env jupyter biopython matplotlib`
- Activate the environment with: `conda activate bio_env`
- Start a Jupyter Notebook kernel with: `jupyter notebook`



(base) francoaquistapace@192 ~ %

A red arrow points to the text "(base)" in the terminal prompt.

How to set up the practical in Google Colab

Disclaimer: you will need to perform these steps every time the Google Colab runtime is disconnected

- Go to <https://colab.research.google.com/> (you may need to use your personal Google account)
- Select the option *Upload* and browse for the notebook you want to upload from your local machine, i.e. our practical session
- Uncomment the installation line and run it (this may take some time):

```
▶ # Uncomment the following line if you need to install dependencies  
!pip install biopython matplotlib  
# You should have biopython >= 1.83
```

How to set up the practical in Google Colab

Disclaimer: you will need to perform these steps every time the Google Colab runtime is disconnected

- Once the runtime has been connected, you need to create a Data folder and upload our files:

